

Weekly CT-01 Answer Sheet

Cambridge International AS and A Level Computer Science 9618

Data Representation Multimedia Representation and Compression

Maximum Marks	20
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Award marks for equivalent wording that communicates the same technical point.

- 1. Bitmap terms** [2]
- Pixel: a single square of one colour, or the smallest addressable element in an image. 1
 - File header: data about the bitmap image, such as file format, image dimensions, colour depth, or file size. 1
- 2. Bitmap file size** [3]
- $1024 \times 512 = 524\,288$ pixels. 1
 - $524\,288 \times 8 = 4\,194\,304$ bits = 524 288 bytes. 1
 - $524\,288 / 1024 / 1024 = 0.50$ MiB. 1
- 3. Lossless bitmap compression** [2]
- Names run-length encoding or RLE. 1
 - Identifies a sequence of consecutive pixels with the same colour. 1
 - Stores the colour value once together with the number of times it repeats. 1
- 4. ASCII** [4]
- ASCII represents 128 characters, or 2^7 . 1
 - Extended ASCII represents 256 characters, or 2^8 . 1
 - Each character has a unique ASCII code; each letter in HOUSE is replaced by its code. 1
 - The codes are stored in the same order as the letters in the word. 1
- 5. Sample rate** [4]
- A higher sample rate creates smaller time gaps between samples, so the digital waveform follows the original more closely or has smaller quantisation error. 2
 - More samples are taken per second, so more data or more bits must be stored, increasing the file size. 2
- 6. Sound compression** [4]
- Compression reduces the file size, making the file faster to transmit or download and more likely to fit an email attachment limit. 2
 - Award up to 2 marks for one valid lossless method with description:
 - Run-length encoding stores one sound value and its repeat count when consecutive samples are identical.
 - Alternatively, store changes between samples instead of every complete sample value
- 7. Atleast 2 hours of mindful study is expected** [0.5]
- 8. Thank you again** [0.5]

Total = 20 mark